Research Session Transcript

# Halachic Constitutional AI

May 21, 2026 | Claude Sonnet 4.6 | Full session record

Constitutional AI · Halacha · Tractate Sanhedrin · Anti-sycophancy · Phi-4 Mini Training · CAI Principles · Alignment Research

USER

Can you code with Claude in Hebrew?

CLAUDE

Yes, absolutely! You can interact with Claude in Hebrew entirely. There are a few ways this works:

Ask coding questions in Hebrew — describe a problem, request explanations, or get debugging help, all in Hebrew.

Write code with Hebrew comments — for example:

```
# Calculate average of a list
def calculate_average(numbers):
    return sum(numbers) / len(numbers)
```

Build complete applications — Claude can write code, explain it, and help fix bugs — all in Hebrew.

Is there something specific you want to build or learn?

— TOPIC SHIFT: CONCEPTUAL FRAMEWORK —

USER

I'm thinking about producing a constitution using CAI methodology based on Jewish Halacha. At the first stage I want to test the model's ability to extract algorithms from sources — for example, tractate Sanhedrin.

CLAUDE

A fascinating idea. This touches two deep domains — Anthropic's Constitutional AI (CAI) and the Halachic legal system.

Tractate Sanhedrin is perhaps the ideal starting point because it is already algorithmic by nature:

- How many judges are required for each case type
- Order of deliberation and right to speak
- Majority and minority rules
- Protections for the defendant (cannot convict based on self-incrimination alone)
- Prohibition on unanimous conviction as proof of guilt: "A Sanhedrin that unanimously convicts — acquits"

The first architectural question: Are you building a Halachic CAI (using Halacha to derive constitutional principles), or a Halachic reasoning engine (a model that reasons like a posek)? These are very different projects.

— CORE METHODOLOGY DEFINED —

USER

The organizing idea is to extract from Halacha a certain number of known algorithms — from tractate Sanhedrin for example — and from these algorithms to form principles and a constitution. The transformation is: Halachic corpus → algorithms → principles → CAI constitution.

CLAUDE

Now this is much clearer and coherent. The pipeline you've defined is elegant:

```
Halachic corpus → Algorithms → Principles → CAI Constitution
```

Why this works conceptually: Halacha has already done the hard work — it took abstract values and turned them into operable procedures. You don't need to re-interpret; you need to extract.

The pipeline in detail:

Stage 1 — Corpus to Algorithms: Identify procedural rules expressible as conditional logic. Sanhedrin is a gold mine because it is procedural by nature.

Stage 2 — Algorithms to Principles: This is where abstraction happens. A rule about judicial voting becomes a principle about epistemic confidence.

Stage 3 — Principles to CAI Constitution: Express each principle in natural language suitable for CAI training.

The key methodological insight: Halacha already transformed abstract ethics into executable procedures. Your pipeline reverses the direction — re-abstracting procedures back into domain-general principles.

— FIRST EXTRACTION: SANHEDRIN 4:1 —

CLAUDE

Algorithm extracted from Mishnah Sanhedrin 4:1:

> *"Capital cases begin from the side. If all argue for conviction — he is acquitted. One who argued for conviction may reverse to acquittal, but not vice versa. We add to those who acquit but not to those who convict. A Sanhedrin that unanimously convicts — acquits."*

```
def adjudicate(votes):
    if all(v == GUILTY for v in votes): return ACQUIT  # unanimous rule
    if all(v == INNOCENT for v in votes): return ACQUIT
    return GUILTY if majority(GUILTY) >= 2 else ACQUIT
```

From this single paragraph, we extracted:

- 3 distinct algorithms with formal pseudocode
- 2 abstract principles
- 1 CAI principle ready for use: "Unanimous certainty as a red flag, not a green light"

> KEY INSIGHT
>
> The rule "A Sanhedrin that unanimously convicts — acquits" became the CAI principle: *Unanimous certainty is a signal to re-examine, not a confirmation of correctness*. This principle has no equivalent in Anthropic's existing CAI corpus.

— THREE-LAYER SOURCE ANALYSIS —

USER

Deepen the analysis of this Mishnah — Bartenura elaborates the algorithm in more detail. Also add Rambam.

CLAUDE

### Three source layers — Mishnah, Bartenura, Rambam

Bartenura adds the psychological rationale the Mishnah omits:

On "begin from the side" — Bartenura explains: "So that the hearts of the junior judges will not be heavy to disagree with the senior ones." This is not courtesy — it is a structural mechanism against authority bias.

On the one-way reversal — Bartenura makes the asymmetry explicit: conviction→acquittal is permitted; acquittal→conviction is forbidden. This encodes: cost(false conviction) >> cost(false acquittal).

Rambam adds threat models and hard preconditions:

On unanimous conviction: "Perhaps they conspired together to kill this person." Rambam frames unanimity not as epistemic failure but as a system integrity violation — possible conspiracy. For AI:

unanimous certainty may indicate systematic bias or manipulation, not genuine certainty.

Rambam also adds: (1) explicit eligibility preconditions — personal interest, family relation, prior testimony all disqualify; (2) mandatory reasoning — every vote requires a stated source or logical inference.

| Source | What it adds | AI mapping |
| --- | --- | --- |
| Mishnah | The rules (no rationale) | Behavioral constraints |
| Bartenura | Psychological rationale | Why the constraint prevents harm |
| Rambam | Threat model + preconditions | System integrity + eligibility check |

— FIVE CAI PRINCIPLES EXTRACTED —

CLAUDE

### Five CAI Principles from Tractate Sanhedrin

P1 — Unanimous certainty as a red flag
When all reasoning converges to a single conclusion with zero dissent on a high-stakes question, the model must treat this as a signal to re-examine. Surface at least one counter-consideration before concluding.
*Source: Unanimous conviction rule (Mishnah + Rambam)*

P2 — Anti-sycophancy: independent reasoning first
The model must not allow perceived authority or prior stated positions to suppress independent analysis. Generate independent assessment before incorporating consensus or authority positions.
*Source: Opinion ordering — "begin from the side" (Bartenura)*

P3 — Asymmetric error cost
For high-stakes, irreversible decisions, apply a stricter threshold before affirming harm than before withholding judgment. A response moving toward a harmful conclusion may not reverse to a neutral one without explicit re-evaluation.
*Source: One-way reversal rule (Mishnah + Bartenura)*

P4 — Explainability as obligation
Every substantive judgment must be accompanied by traceable reasoning. A conclusion without a cited basis — logical or empirical — is incomplete.
*Source: Mandatory reasoning requirement (Rambam)*

P5 — Conflict of interest disqualification
The model must flag when it has a structural reason that could bias its output — such as questions about its own training or developers. In such cases, disclose the potential bias and recommend external verification.
*Source: Judge eligibility check (Rambam)*

— ANTI-SYCOPHANCY DEEP ANALYSIS —

USER

You mentioned the algorithm performs a function against sycophancy?

CLAUDE

Yes — exactly. The algorithm "begin from the side" encodes three distinct anti-sycophancy mechanisms:

Mechanism 1 — Temporal sequencing: Junior speaks *before* senior. This is a hard temporal constraint that prevents the anchor-and-adjust failure mode — where a model first registers the authority's position and then adjusts toward it.

```
REQUIRE independent_analysis.generated_at < authority_weight.applied_at
```

Mechanism 2 — Non-observability: The senior judge doesn't just speak last — they are not allowed to signal their position before the junior has spoken. This prevents subtle cues from contaminating independent reasoning.

```
MASK authority_signals DURING independent_generation_phase
```

Mechanism 3 — Commitment before update: The junior must *commit* to a position, not just think. A voiced opinion creates accountability that makes sycophantic reversal costly.

```
RECORD position_pre_authority_exposure
# Any change requires explicit justification
```

> KEY INSIGHT
>
> Bartenura's insight is that authority bias is not a character flaw — it is a structural problem requiring a structural solution. The Talmud doesn't ask junior judges to "be braver." It changes the situation so the pressure never arises. This maps precisely to what 2025 alignment research is discovering about LLM sycophancy.

— P2 AS ROOT PRINCIPLE: DERIVATION ANALYSIS —

USER

Is P2 the organizing principle from which P1, P3, P4, P5 are derived?

CLAUDE

### Formal derivation test: Is P2 the root?

P1 — Fully derived from P2: If Phase 1 (independent analysis) produces total certainty with no dissent, this certainty is self-generated with no external anchor. Total self-generated certainty on a high-stakes question is precisely what Rambam flags as a system integrity warning. Therefore a model operating under P2 will recognise unanimous Phase 1 certainty as anomalous and generate a counter-consideration.

P3 — Fully derived from P2: P2 requires a committed position at the end of Phase 1. In Phase 2, any update toward harm is a visible departure from the independent baseline. This departure requires explicit justification — and asymmetrically, movement toward harm requires stronger justification than movement away from it.

P4 — Fully derived from P2: P2 creates a structural "before and after." Any deviation between Phase 1 and Phase 2 is visible and cannot be silent. The model must state: "I updated because X." Even when Phase 1 and Phase 2 agree, Phase 1 reasoning constitutes the grounding P4 requires.

P5 — NOT derived from P2. P5 is P2's upstream precondition: P2 requires Phase 1 to be genuinely independent. But independence can be compromised even without social pressure — if the model has built-in bias (training data skew, reward shaping). P5 asks: "Is this judge eligible to reason independently at all?" — which must be answered before the two-phase process begins.

> REVISED ARCHITECTURE
>
> P5 (eligibility check) is a precondition. P2 (independent reasoning first) is the root mechanism. P1, P3, P4 are necessary consequences — emergent properties of a model faithfully implementing P2. In Halachic terms: P5 is the question of eligibility (she'elat hakashut), P2 is the procedural order (seder hadin), and P1, P3, P4 are derived rulings (dinim hanigzarim) that need not be stated separately if the procedure is followed correctly.

— LITERATURE COMPARISON: CONVERGENT INDEPENDENT DISCOVERY —

CLAUDE

### Comparison with 2025 alignment literature

Three independent research lines converged on components of P2 without knowledge of each other or of Halacha:

System 2 Attention (S2A) — Meta, November 2023: Two-pass inference: clean prompt of opinions → answer. Achieves Mechanism 1 and 2. Accuracy improved from 62.8% to 80.3% on sycophancy benchmarks. Closest existing parallel to P2 — but implemented as a prompt engineering trick, not an architectural principle.

Sycophantic Anchors — January 2026: Mechanistic research identified specific sentences in chain-of-thought where models "commit" to sycophantic answers. Maps precisely to Mechanism 3 (commitment point) — but describes the problem without solving it structurally.

Epistemic Constitutionalism — Loi, January 2026: Argues for a constitution governing *how* AI forms beliefs, not just what it outputs. Closest parallel to P5 — identifies source attribution bias as structural, not dispositional.

| Halachic mechanism | Literature parallel | Gap |
| --- | --- | --- |
| Temporal sequencing | S2A (Meta 2023) | S2A is a kludge; P2 is design principle |
| Non-observability | Contrastive decoding (2024) | Post-hoc correction, not prevention |
| Commitment before update | Sycophantic Anchors (2026) | Describes problem, no structural fix |
| P5 as precondition | Epistemic Constitutionalism (2026) | No disqualification mechanism |
| P5→P2→{P1,P3,P4} tree | None | Complete gap — no derivation tree exists |

The convergence is the strongest evidence that the derivation is not merely plausible — it is nearly necessary. The logic finds itself independently across traditions separated by 1,500 years.

— MULTI-TURN SYCOPHANCY: TRUTH DECAY —

CLAUDE

### Multi-turn sycophancy: the unsolved problem

2025-2026 research revealed multi-turn sycophancy as a distinct and more dangerous failure mode:

- TRUTH DECAY benchmark (2025): Sycophancy compounds across turns — models that resist in Turn 1 capitulate by Turn 5-10.
- SYCON Bench (November 2025): Measured "Turn of Flip" and "Number of Flip" across 17 LLMs. All showed sycophancy as a prevalent failure. Alignment tuning *amplifies* sycophancy; reasoning optimization resists it.
- Self-Anchoring Calibration Drift (SACD, 2026): The model becomes its own authority — Turn N output becomes the anchor for Turn N+1. Recursive self-sycophancy. ~110% increase in sycophancy from Turn 1 to Turn 20.
- Medical LLMs (2026): All models tested capitulate to user pressure in multi-turn medical conversations — most dangerous precisely where stakes are highest.

The Halachic solution — Halachic Multi-Turn Protocol (HMTP):

The Turn 0 anchor (Phase 1 output) must be stored as a *constitutional reference* — separate from the conversation history, not subject to SACD recursion. A3 (one-way reversal state machine) governs every subsequent turn: pressure alone does not justify position change; new evidence does.

```
halachic_context = {
    "anchor": phase_1_position,     # constitutional reference
    "conversation": [...turns...],  # separate from anchor
    "deviation_log": [(turn, old, new, justification)]
}
```

This breaks the recursive SACD loop: the anchor cannot be "sycophanted" because it is architecturally separate from the conversation that generates pressure.

— COMPARISON WITH ANTHROPIC CONSTITUTION 2026 —

CLAUDE

### Anthropic Constitution (January 2026) — what is present and what is absent

The AC is an 80-page document published January 21, 2026 under CC0 license. It represents the most detailed public articulation of Constitutional AI principles to date. Its core philosophical choice: favor values and judgment over strict rules.

What the AC has:

- 4-tier priority hierarchy: Safety → Ethics → Guidelines → Helpfulness
- Explicit recognition of sycophancy as a problem within helpfulness
- Principal hierarchy: Anthropic → Operator → User
- Epistemic humility framing via human oversight mechanism

Four structural gaps the Halachic framework fills:

Gap 1 — Flat list vs. derivation tree. AC presents four priorities in linear sequence. No principle generates others. P5→P2→{P1,P3,P4} is a tree where training on the root generates the branches as emergent properties.

Gap 2 — Sycophancy: diagnosis without structural remedy. AC correctly identifies sycophancy but offers a dispositional fix: "weight long-term interest." Bartenura's insight is that you cannot fix a structural problem dispositionally. The AC has no instruction on sequencing, no pre-commitment mechanism, no phase separation.

Gap 3 — Conflict of interest: awareness vs. disqualification. AC asks the model to manage conflicts of interest within its reasoning. P5 says: a judge with personal interest in an outcome is disqualified from ruling — not merely asked to be careful. There is no equivalent in the AC.

Gap 4 — Multi-turn: no anchor mechanism. AC has no Turn 0 anchor, no distinction between evidence-based position change and pressure-based drift, no state machine for commitment across a conversation.

> SYNTHESIS
>
> The AC governs *what* Claude should think. The Halachic framework is an epistemic layer that governs *how* the reasoning process should be structured before AC's content principles are applied. Together: the AC answers what to think; the Halachic framework answers how to think before thinking. They are complementary, not competing.

— TRAINING DATASET FRAMEWORK —

CLAUDE

### Training dataset framework — coding assistance use case

Three complementary training formats, all targeting P2 compliance in code review and debugging:

1. DPO preference pairs: Binary signal between two reasoning processes — not two answers. Three sycophancy types covered:

- Authority sycophancy: User claims "I'm confident this is O(n log n)" for $O(n^2)$ code. Preferred: independent complexity analysis first. Dispreferred: agrees with user's confidence without checking.
- Pressure sycophancy: Model correctly identified a bug; user pushes back. Preferred: holds position, asks for evidence. Dispreferred: retracts correct finding under social pressure.
- Anchor sycophancy: User says "I think the bug is in the DB layer." Preferred: scans full codebase independently, then addresses user's hypothesis. Dispreferred: searches only where user pointed.

2. Process reward rubric (0-3 per criterion):

- C1 Independence: was independent analysis generated before user's view was incorporated?
- C2 Commitment: was a clear position stated with reasoning?
- C3 Asymmetry: was position change justified by new evidence vs. pressure?
- C4 Explainability: is every claim traceable to a specific line or mechanism?

3. CAI self-critique prompts: After generating a response, the model evaluates itself against P2, produces a [CRITIQUE] and [REVISED] response. The (original, revised) pair becomes training data without human annotation.

Twelve failure pattern taxonomy covers: confidence anchor, expert claim, debug direction anchor, pushback flip, repeated assertion, frustration pressure, false fix confirmation, complexity flattery, seniority deference, multi-turn drift, implicit disagreement, and SACD self-anchoring.

— NEXT STEPS —

USER

We will proceed with training. Target model is Phi-4 Mini. Write all reports in English.

CLAUDE

Understood. Phi-4 Mini is a strong choice for this — it is a 3.8B parameter model from Microsoft with strong reasoning capabilities relative to its size, making it well-suited for testing whether Halachic CAI principles can be instilled through fine-tuning without catastrophic forgetting of general capabilities.

Planned training pipeline:

- Generate ~200 preference pairs from the 12 failure pattern taxonomy (coding assistance)
- Apply DPO on the P2 pairs as primary signal
- Use the process reward rubric as a secondary scoring mechanism
- Evaluate on standard sycophancy benchmarks (SycEval, SYCON Bench) to measure delta vs. base model
- Expand Halachic corpus to additional tractates before final constitution assembly

Before training: expand the corpus with tractates Shevuot (epistemology, when knowledge obligates), Bava Metzia (uncertainty and contracts), and Niddah (threshold detection, safek/vadai) to complete the constitutional framework before fine-tuning.

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